

REPORT

ASSIGNMENT 6

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INTRODUCTION

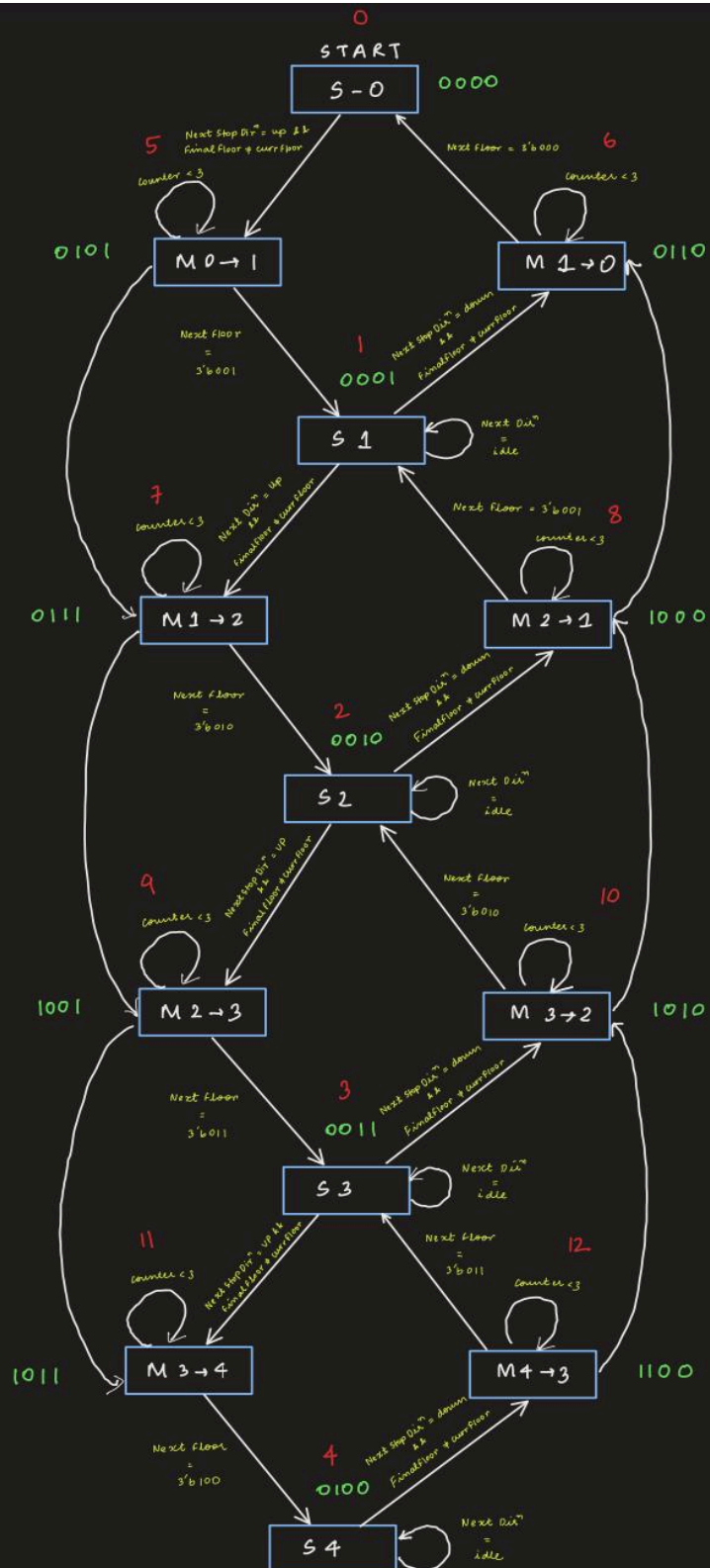
The objective of this assignment is to design and implement a lift controller using Verilog and demonstrate its functionality on an FPGA (Field Programmable Gate Array). The lift controller is tasked with managing the operation of a lift system serving four floors with five stoppages, including the ground floor.

The problem statement outlines several key requirements for the lift controller:

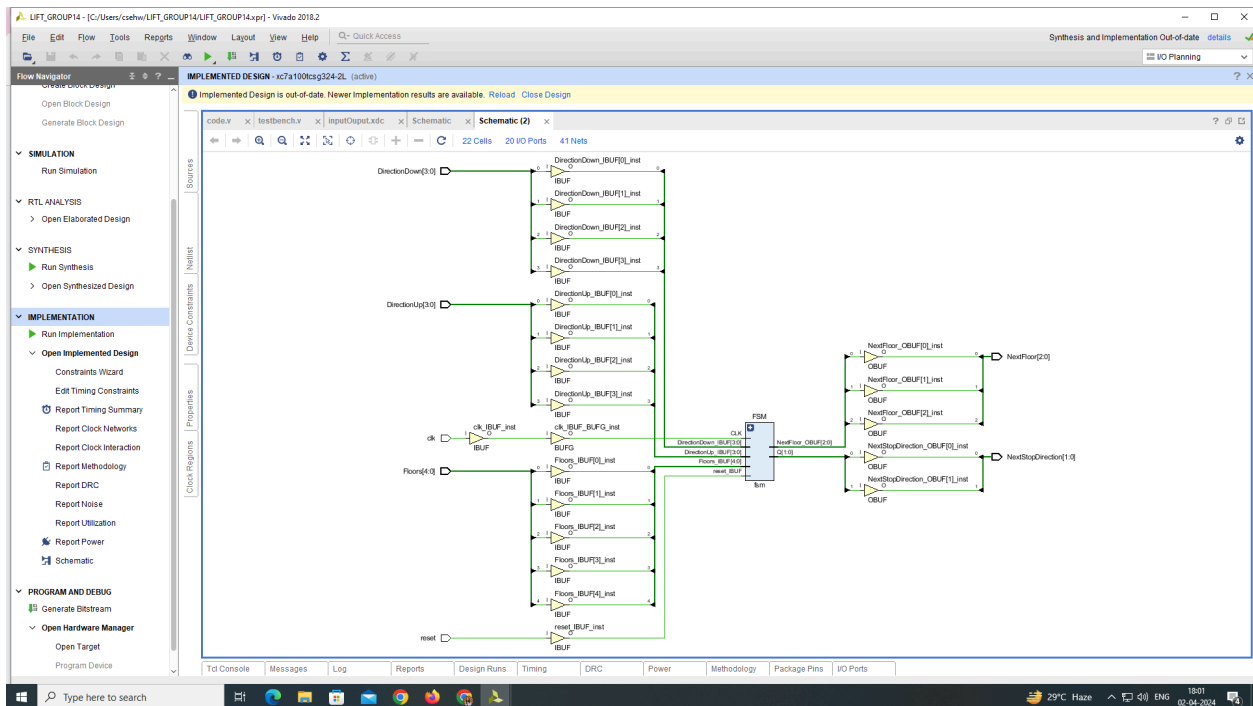
- **Basic Functionality:** The lift controller must prioritize serving the current request first and accommodate additional requests if possible without impacting the current operation. Requests can come from any floor for the lift to move up or down.
- **Dynamic Next-Stop Determination:** The next floor the lift will travel to must be dynamically determined based on all inputs at the current clock cycle. The lift's direction (up or down) for the next clock cycle is also determined based on these inputs.

- **Handling Multiple Requests:** Multiple passengers may enter the lift from the same floor, resulting in multiple floor requests in a single clock cycle. The controller must handle these requests efficiently.
- **Directional Priority:** When multiple requests are received simultaneously, the lift must prioritise the nearest request in the direction it is currently moving. If idle, it should move downward.
- **Clock Cycle Timing:** The lift takes four clock cycles to move between adjacent floors. Requests can be received in every clock cycle, and the lift's next stop dynamically changes based on these requests.

FSM



SCHEMATIC



SIMULATION

